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Simplifying Scientific Text

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1. Chapter One

1.1 Introduction

Natural Language Processing has developed significantly with the emergence of large language models, which are now able to understand, rewrite, and generate text in a flexible and coherent manner. Text simplification is one of the important applications in this field, as it aims to transform complex text into a clearer and easier form while preserving the original meaning.

The need for text simplification becomes more important in scientific and medical domains, where texts often contain specialized terminology, long sentence structures, and detailed information that may be difficult for non-specialist readers to understand. This project focuses on building an intelligent system for simplifying scientific and medical sentences using large language models, with the goal of making information more accessible without changing the essential meaning.

1.2 Problem of Project

The scientific problem addressed in this project is the difficulty of understanding scientific and medical texts by non-specialist readers. Such texts often include complex terminology, long grammatical structures, and domain-specific details. Although large language models can rewrite text, using them directly may sometimes lead to removing important information, changing the meaning, or adding unsupported content.

Therefore, there is a need for an intelligent simplification system that does not process all sentences in the same way. Instead, the system should determine the suitable simplification operation for each sentence, such as rephrasing, splitting, omitting unnecessary details, substituting difficult terms, or keeping the sentence unchanged, and then generate a simpler version while preserving the core scientific meaning.

1.3 Project Objectives

The main objective of this research is to build an intelligent system that helps make scientific and medical sentences easier to understand for non-specialist readers while preserving the original meaning of the source text. The research aims to reduce the gap between complex scientific writing and general readers by improving the clarity and readability of specialized texts, especially those that contain difficult terminology, long sentence structures, and dense information. It also seeks to support accurate access to scientific and medical knowledge by simplifying the language without adding unsupported information, removing essential content, or changing the intended meaning.

1.4 Project Contribution

The main contribution of this project is the development of a structured text simplification system that combines operation selection with large-language-model-based generation. Instead of applying a single fixed rewriting prompt to all sentences, the proposed system first analyzes the sentence and determines the appropriate type of simplification, then generates the simplified output according to that decision.

The project also contributes by combining several components within one pipeline: large language models, operation-aware planning, and optional medical term support. This design allows the system to handle different types of scientific sentences, including long sentences, terminology-heavy sentences, and sentences that contain secondary details. In addition, the project provides an experimental evaluation of the proposed approach using both quantitative metrics and qualitative analysis.

1.5 Beneficiaries of the Project

Several groups can benefit from this project, including:

- Non-specialist readers who need to understand scientific and medical information more easily.
- Patients and caregivers who may benefit from clearer medical explanations.
- Students and early-stage projecters who need support in understanding scientific summaries and abstracts.
- Educational institutions that aim to prepare simplified scientific learning materials.
- Health and educational platforms that provide accessible content for the general public.
- Projecters in Natural Language Processing who study text simplification and large language model applications.

1.6 Thesis Organization

This thesis is organized into six chapters. Chapter One introduces the project topic, the scientific problem, the project objectives, the contribution, and the expected beneficiaries. Chapter Two presents the theoretical background related to Natural Language Processing, text simplification, large language models, and evaluation metrics.

Chapter Three reviews related work and previous studies in scientific and medical text simplification, with a discussion of the project gap addressed by this project. Chapter Four explains the proposed methodology, including the system architecture, dataset, preprocessing steps, operation selection, generation process, and experimental setup. Chapter Five presents the evaluation results and compares the proposed strategies. Finally, Chapter Six summarizes the main conclusions, discusses the limitations, and suggests future improvements.

2.Chapter Two: Theoretical Background

This chapter presents the theoretical and computational concepts required to understand the proposed project.

2.1 Natural Language Processing

Natural Language Processing (NLP) is a branch of artificial intelligence concerned with enabling computers to process, analyze, understand, and generate human language. NLP is used in many applications such as machine translation, text classification, information extraction, summarization, question answering, and text simplification.

In this project, NLP is the main computational field because the system receives textual input and produces a modified textual output. The goal is not only to process the sentence structurally, but also to generate a simpler version that remains faithful to the original meaning.

2.2 Text Simplification

Text simplification is the task of rewriting a complex text into a clearer and more readable form while preserving its essential meaning. The objective is to reduce linguistic complexity without changing the core information expressed in the original text.

Simplification may involve several linguistic changes, such as replacing difficult words with simpler alternatives, rewriting complex expressions, shortening long constructions, or restructuring a sentence to make it easier to read. In some cases, the best output may be very close to the original sentence if the sentence is already understandable.

In this project, text simplification is addressed at the sentence level. Each input sentence is treated as an independent unit that should be simplified according to its content and linguistic structure.

2.3 Neural Language Models

Neural language models are computational models that learn patterns of language from large text datasets. Instead of relying only on manually written linguistic rules, these models learn statistical and semantic relationships between words and phrases through training.

A language model estimates the likelihood of word sequences and uses this knowledge to generate or transform text. Modern neural language models represent words or tokens as numerical vectors, allowing the model to capture relationships between linguistic units in a continuous mathematical space.

These models form the basis of many modern NLP systems because they can learn contextual information and produce fluent text. However, their output still depends strongly on the training data, model architecture, and instructions provided during inference.

2.4 Transformer Architecture

The Transformer is one of the most important architectures used in modern NLP. It was designed to process sequential data, especially natural language, using an attention mechanism instead of relying only on recurrent processing.

The main idea of the Transformer is self-attention. Self-attention allows the model to focus on different parts of the input sequence when representing or generating each token. This enables the model to capture long-range dependencies and contextual relationships more effectively than many earlier architectures.

A Transformer model usually includes several core components, such as token embeddings, positional information, attention layers, and feed-forward neural networks. These components allow the model to transform an input sequence into contextual representations that can be used for understanding or generation tasks.

2.5 Large Language Models

Large Language Models (LLMs) are advanced neural language models trained on very large text corpora. They are capable of generating coherent text, answering questions, summarizing content, following instructions, and performing several NLP tasks without task-specific training in some cases.

LLMs are useful for text simplification because they can rewrite complex sentences in simpler language. However, their use must be controlled carefully. A model may sometimes simplify too aggressively, remove important information, or generate unsupported content. Therefore, the quality of the output depends on the model, the prompt, the constraints, and the evaluation process.

Examples of large language model families that may be considered in this project include LLaMA, Qwen, Gemini, Gemma, and other instruction-tuned models. The final selection depends on experimental results, computational feasibility, cost, and output quality.

2.6 Prompt Engineering

Prompt engineering is the process of designing clear and effective instructions for a language model. Since large language models respond according to the input prompt, the structure and wording of the prompt have a direct effect on the generated output.

In text simplification, prompt engineering is important because the model must be guided to produce simpler text while preserving the original meaning. A good prompt should define the task, specify the expected behavior, and include constraints such as avoiding unsupported information, preserving important details, and using clear language.

Prompt design can also help improve consistency, especially when the system requires outputs in a specific format or when different types of sentence transformations are needed.

2.7 Terminology Support in Scientific Text Simplification

Scientific texts often contain specialized terms that may be difficult to simplify correctly. To support the simplification process, a system may use a terminology-support component, such as a glossary of domain-specific terms and short definitions.

In this project, terminology support is considered as an auxiliary component rather than a full retrieval-based question answering system. Its role is to provide limited contextual hints about difficult scientific or medical terms when they appear in the input sentence. These hints can help the language model understand the meaning of specialized terms and choose clearer wording.

This component should not be used to add new information that is not present in the original sentence. Instead, it should support faithful simplification by helping the model preserve meaning while making the sentence easier to understand.

3. Chapter Three: The CLEF SimpleText Shared Task

This project is developed within the context of the CLEF SimpleText shared task, which focuses on making scientific text easier to understand using Natural Language Processing methods. The task provides a common evaluation setting where systems can be compared using shared data and evaluation metrics. SimpleText includes simplification at both the sentence level and the document level; however, this project focuses on Task 1.1: sentence-level scientific text simplification, where the system receives a complex scientific or medical sentence and generates a simpler version while preserving the original meaning. The task uses data derived from Cochrane-auto, which contains aligned scientific abstracts and plain-language summaries. In this project, the sentence-level part is considered at the current stage, while the dataset structure, preprocessing steps, and data splits will be described in detail in the methodology chapter.

4. Chapter Four: Related Work

4.1 Introduction

This chapter reviews the main research papers presented in the **SimpleText 2025** lab. The reviewed works are organized according to the technical methodologies they used for scientific and medical text simplification. These methodologies include zero-shot prompting and prompt engineering, fine-tuning and supervised learning, hallucination detection, plan-driven simplification, and lexical or terminology-supported simplification.

The purpose of this review is to clarify the main research directions in the field, highlight the strengths and limitations of each methodology, and position the current project within these related works. In particular, the current project is related to approaches that combine operation-aware planning, large language models, and limited terminology support for controlled sentence-level simplification.

4.2 Zero-Shot Prompting and Prompt Engineering Approaches

A major group of SimpleText 2025 systems relied on zero-shot prompting and prompt engineering. These approaches attempt to use the existing capabilities of large language models without additional task-specific training. Their performance depends mainly on the quality of the prompt, the selected model, and the constraints imposed during generation.

The LIA team from Avignon used human-inspired simplification guidelines derived from Cochrane recommendations to guide models such as Mistral and LLaMA in document simplification. Their best system, referred to as Guidelines-all, achieved a SARI score of 44.93, indicating strong performance among prompt-based systems. This approach highlights the importance of providing the model with explicit simplification principles rather than relying on a general rewriting instruction.

The SINAI team used GPT-4.1 for medical text simplification, with particular attention to explaining difficult terms between parentheses. Their system achieved a SARI score of 41.25 in Task 1.1 and 43.63 in Task 1.2. This work shows that prompt engineering can be effective when the model is instructed to handle terminology explicitly.

The UBOnlp team explored the raw simplification ability of GPT-4o using a simple prompt. Their method focused on reducing content and generating simpler text. The system achieved a SARI score of 42.20 at the sentence level and 43.37 at the document level. Although the approach was relatively simple, the results suggest that strong general-purpose LLMs can perform competitively even with limited task-specific engineering.

The Fujitsu Lenguaje-Claro team compared zero-shot prompting with three-shot prompting, where the examples were designed to represent specific linguistic phenomena. Their GPT-3.5-Turbo three-shot configuration achieved a SARI score of 38.84. This result suggests that few-

shot prompting may improve control in some cases, although its effectiveness depends on the quality and representativeness of the selected examples.

4.3 Fine-Tuning and Supervised Learning Approaches

Another group of systems relied on fine-tuning or supervised learning. These approaches train or adapt models directly on simplification datasets, such as Cochrane-auto, in order to improve task-specific performance.

The UM_FHS team compared prompt-based methods with fine-tuned versions of GPT-4.1 mini and GPT-4.1 nano. Their gpt-4.1-mini without context achieved a SARI score of 42.13. This result shows that smaller instruction-tuned models can achieve competitive performance when appropriately adapted or prompted.

The NITK SCaLAR team fine-tuned GPT-2 Medium for direct simplification and also proposed a hybrid system combining BioBERT and BioBART. Their GPT-2 Medium system achieved a SARI score of 39.77. This work reflects the use of supervised neural models for simplification, although its performance remained below the strongest prompt-engineering systems.

The AIIRLab team used Unsloth to fine-tune quantized versions of Mistral and LLaMA models. Their Mistral-7B system achieved a SARI score of 36.08 at the sentence level and 42.40 at the document level. This approach is relevant because it shows how quantized models can be adapted for simplification under computational constraints.

The DUTH team applied instruction tuning to models such as FLAN-T5-Large and BART-SAMSum. Their FLAN-T5-Large model achieved a SARI score of 35.35. Although the result was lower than several LLM-based systems, this approach remains important because encoder-decoder models such as T5 are naturally suitable for sequence-to-sequence tasks like simplification.

In general, fine-tuning approaches offer better task specialization, but their performance depends strongly on the training data, model size, and available computational resources. In comparison, prompt-based LLM systems achieved higher scores in several cases without requiring full model training.

4.4 Hallucination Detection Approaches

Hallucination detection was mainly associated with the second task of SimpleText 2025, which focused on identifying unsupported or inaccurate information in simplified texts. This direction is important because simplification systems may produce fluent text while changing or adding information that is not supported by the source.

The SINAI team combined rule-based pattern detection, such as identifying single-word sentences, with LLaMA-3.1. Their best system achieved an F1-score of 0.976 for source-based hallucination detection. This result indicates that combining simple rules with large language models can be highly effective for detecting unsupported content.

The DS@GT team developed a hybrid hallucination detection system that combined BERT classifiers, Natural Language Inference models, and LLMs. Their system achieved an F1-score of 0.95 when the source was available. This approach shows the value of combining classification and inference-based methods to verify whether a simplified output remains faithful to its source.

The DUTH team used tree-based classifiers, particularly Extra Trees, to analyze lexical and semantic features. Their system achieved an F1-score of 0.950 for hallucination detection. This result shows that traditional machine learning methods can still be useful when supported by carefully designed features.

4.5 Plan-Driven and Structural Simplification Approaches

Plan-driven and structural approaches attempt to decompose the simplification process into explicit steps. Instead of applying a single rewriting instruction to all sentences, these systems identify a simplification strategy or structural transformation before generating the final output.

The DS@GT team, which achieved the highest sentence-level performance in Task 1, used a plan-guided system in which the model selected a strategy such as deletion or splitting before generating the simplified text. Their system achieved a SARI score of 42.98. This result is especially relevant to the current project because it supports the idea that operation-aware simplification can improve control over sentence transformations.

The PICT team proposed the S-3 Pipeline, which combined structural AMR representations with semantic simplification using T5 and lexical simplification using BERT-Masking. Their system achieved a SARI score of 17.77. The team interpreted the low score in the medical context as potentially reflecting stronger meaning preservation. This approach is important because it shows an attempt to use deeper semantic structure rather than relying only on surface-level rewriting.

The UvA team from Amsterdam applied a Plan-guided BART system that identifies the required action for each sentence before processing it. Their system achieved a SARI score of 38.5 in the sentence-level task. This work further supports the relevance of planning in text simplification, especially when different sentences require different transformation operations.

4.6 Retrieval-Augmented and Lexical Simplification Approaches

Some systems focused on lexical simplification and external terminology support. These approaches are particularly important in scientific and medical text simplification because difficult terminology is one of the main barriers for non-specialist readers.

The LIS team introduced a notable contribution through the MedSimplify glossary, which contains approximately 3000 definitions. They used this resource within a retrieval-augmented architecture. Their Mistral_DASP_0 system achieved a SARI score of 43.51. This result indicates that terminology support can improve simplification by helping the model handle medical or scientific expressions more effectively.

The THM team focused on Complex Word Identification. Their system identified complex terms and marked them with square brackets so that the model would simplify those terms specifically. Using gpt-4.1-nano, the system achieved a SARI score of 39.57. This approach shows that explicitly identifying difficult terms can guide the model toward more targeted simplification.

4.7 Comparative Analysis of Reviewed Approaches

The reviewed systems show that there is no single dominant method for scientific text simplification. Prompt engineering approaches achieved strong results, especially when combined with clear simplification guidelines. Fine-tuning approaches provided task-specific adaptation, but did not always outperform strong LLM prompting. Plan-driven systems showed the importance of selecting a suitable simplification strategy before generation, while lexical and retrieval-supported systems demonstrated the value of terminology handling.

The following table summarizes the main reviewed systems and their reported results.

Team / System	Main Methodology	Models Used	Main Reported Result	Strengths	Limitations
UM-FHS	Prompt engineering and fine-tuning with simplification guidelines	GPT-4.1, GPT-4.1-mini, GPT-4.1-nano	SARI 43.34 in Task 1.1; SARI 43.83 in Task 1.2	Strong prompt design; clear simplification guidelines; competitive results	Closed and costly models; fine-tuning results were not always stable; no explicit planning pipeline
LIA	Cochrane-guideline-based prompting for document simplification	LLaMA, Mistral, Gemma	SARI 44.93 in Task 1.2	Strong use of medical simplification guidelines; best document-level performance	Focused on Task 1.2; large models; no post-generation verification
DS@GT	Plan-driven simplification before generation	LLaMA 3.3 70B	SARI 42.98 in Task 1.1	Planning improves control and interpretability; close to current project direction	Requires multiple model calls; possible aggressive deletion; no explicit verification stage
SINAI	Zero-shot medical prompting and hallucination detection	GPT-4.1, LLaMA-3.1	SARI 41.25–41.82 in Task 1.1; F1 0.976 in hallucination detection	Strong zero-shot performance; explicit handling of terms; strong hallucination detection	Depends on closed large models; zero-shot outputs may be unstable on complex cases

LIS	Terminology-supported simplification using MedSimplify glossary	Mistral 7B, Gemma, Med42	SARI 43.51 in Task 1.2	Domain terminology support; no fine-tuning required; relevant to medical simplification	Exact matching only; limited glossary coverage; mainly document-level
UvA	Fine-tuned BART and jargon-aware prompting	BART, LLaMA 3.1 8B, MedReadMe classifier	SARI 42.31 in Task 1.1	Effective use of Cochrane-auto fine-tuning; jargon-aware prompting	No explicit planning stage; no verification after generation
THM	Complex word identification and targeted lexical simplification	GPT-4.1-nano, Gemini models	SARI 41.32 in Task 1.1	Focused handling of difficult terms; economical model option	Closed models; mainly lexical focus; no full structural pipeline
DUTH	Encoder-decoder simplification and hallucination classification	FLAN-T5, BART-SAMSum, Extra Trees	SARI 38.73 in Task 1.1; F1 0.948 in hallucination detection	Practical smaller models; covers simplification and hallucination detection	Lower SARI than LLM-based systems; no medical guidelines or planning
AIIRLab	Quantized LLMs with cross-encoder verification	Mistral 7B, LLaMA 3.1/3.2, BERT-based cross-encoders	SARI 42.44 with base Mistral; SARI 36.08 after fine-tuning	Open-source medium-size models; verification-oriented design	Cross-encoder adds latency; fine-tuning did not always improve performance
PICT S-3 Pipeline	Hybrid lexical and semantic simplification	SciBERT, AMRlib, PEGASUS, BERT, T5	SARI 40.15 in Task 1.1; SARI 40.29 in Task 1.2	Highly structured and interpretable pipeline; lexical and semantic processing	Complex architecture; lower SARI than top LLM systems; no explicit planning
NITK SCaLAR	Biomedical model pipeline and GPT-2 fine-	BioBERT, BioBART, GPT-2 Medium,	SARI 33.95–39.77 in Task 1.1	Uses biomedical representations; lower computational	Lower performance; older models; no planning or

	tuning	SciSpacy		cost	verification
UBOnlyp	Direct prompting with a general-purpose LLM	GPT-4o	SARI 40.74–42.20 in Task 1.1; SARI 41.56–43.37 in Task 1.2	Strong results with a simple setup	Closed and costly model; limited methodological detail; no structured pipeline

4.8 Research Gap and Motivation for the Current Project

The reviewed works reveal several important observations. First, strong LLMs can achieve competitive simplification performance through prompt engineering, but direct prompting may not always provide enough control over the type of transformation applied to each sentence. Second, fine-tuning can specialize models for the task, but it may require additional computational resources and does not always guarantee better performance than well-designed prompting. Third, plan-driven systems show that identifying the intended simplification operation before generation can improve the structure and controllability of the output. Fourth, terminology support is particularly important in scientific and medical texts, where difficult terms can reduce readability and lead to misunderstanding.

Based on these observations, the current project is motivated by the need for a controlled sentence-level simplification pipeline that combines operation-aware planning with large language model generation and limited terminology support. The goal is not only to produce simpler sentences, but also to preserve scientific meaning, reduce unnecessary rewriting, and handle different sentence types through suitable simplification strategies.

This direction is supported by the reviewed plan-driven and lexical simplification systems, while also addressing the need for a practical and modular pipeline that can be tested, evaluated, and improved across different sentence categories.